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English version

Dock levellers

Rampes ajustables

Ladebrücken

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Central Secretariat: rue de Stassart,36 B-1050 Brussels

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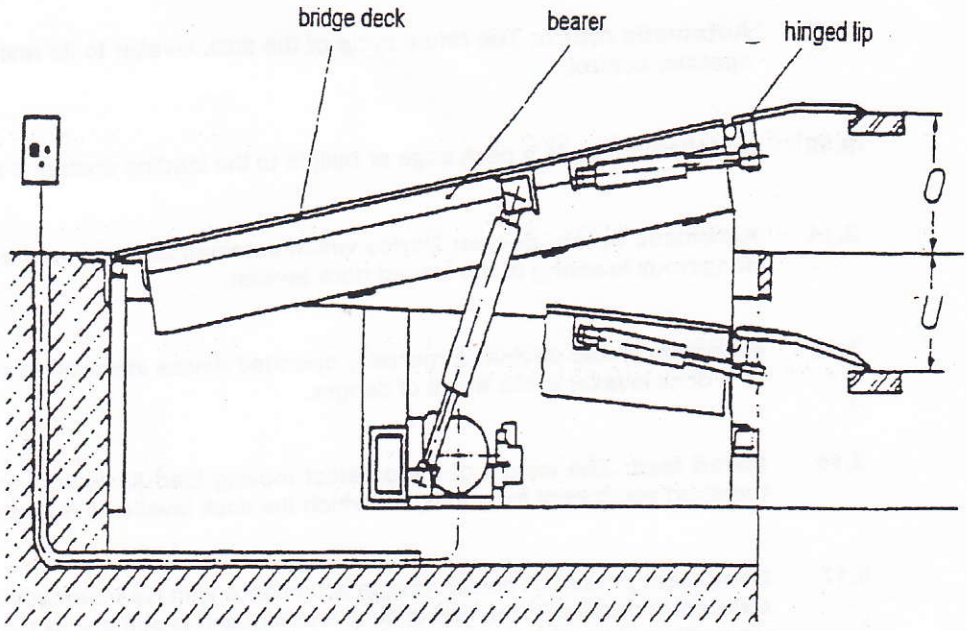


Figure 2: Main parts of a dock leveller

3.5 **Bearer:** Supporting part of a dock leveller in form of a beam, fixed at the under side of the bridge deck (see figure 2).

3.6 **Support device:** Part of a dock leveller, e.g. in form of a block, bar or bolt which supports the dock leveller in the horizontal resting position.

3.7 **Hinged lip:** Part of a dock leveller hinged to the bridge deck and resting on the vehicle in the operational position (see figure 2).

3.8 **Telescopic lip:** Part of a dock leveller which can be telescopically extended towards the vehicle and which rests on the vehicle in operational position.

3.9 **Loading:** Whenever the term "loading" is applied in this standard, it is understood to include loading and unloading.

3.10 **Free floating condition:** A mode in which the dock leveller supported on the vehicle may follow automatically the vertical movements of the vehicle during the loading process.

3.11 **Resting position:** Position to which the dock leveller is put, or to which it returns, when loading has finished.

3.12 **Automatic return:** The return cycle of the dock leveller to its resting position without operator control.

3.13 **Installed:** Fixed to a dock edge or built in to the loading dock or a similar area.

3.14 **Automatic safety device:** Device which automatically prevents an uncontrolled and dangerous lowering of the loaded dock leveller.

3.15 **Emergency stop device:** A manually operated device intended to stop all movements of a dock leveller in the event of danger.

3.16 **Rated load:** The weight of the greatest moving load (including goods, persons and transport equipment for goods) for which the dock leveller is designed.

3.17 **Competent person:** A person who, in accordance with his training and experience, has sufficient knowledge in the field of dock levellers and is sufficiently familiar with relevant regulations to be able to assess the safe condition of dock levellers.

3.18 **Operational position:** Position of the dock leveller in which it is provided for loading and unloading.

4 List of significant hazards

The hazards that can influence the safety of persons during operation, maintenance or inspection have been identified by the risk assessment procedure and the corresponding requirements formulated. Table 1 shows the hazards which have been identified and where the corresponding requirements have been formulated in this standard.

Table 1: Significant hazards and safety requirements

Significant hazards	Safety requirements; Relevant clauses in this standard	Relevant clauses of EN 292-1:1991
Crushing hazard Shearing hazard	6.1.8, 6.1.9, 6.3.2.1, 6.3.2.2, 6.3.2.4, 6.3.2.5, 6.3.2.7, 6.3.2.8	4.2.1
Impact hazard	6.1.10	4.2.1
High pressure fluid ejection hazard	6.3.4.1, 6.3.4.2, 6.3.4.3, 6.3.4.4, 6.3.4.5	4.2.1
Loss of stability	5	4.2.2
Slip, trip and fall hazard	6.1.1, 6.1.2, 6.1.3, 6.1.4, 6.1.5, 6.1.6, 6.1.7, 6.1.10, 6.2.1, 6.3.1	4.2.3
Electrical contact	6.3.2.6, 6.3.3	4.3
Failure of energy supply	6.3.2.7, 6.3.2.8	4.3
Hazards generated by neglecting ergonomic principles in machine design	6.2.2, 6.2.3	4.9
Failure, malfunction of control system	6.3.2.5, 6.3.2.7	4.10

5 Calculations and dimensions

Dock levellers shall be designed to withstand all loads and forces under all operating and static conditions according to the scope.

Dock levellers have an adequate strength if when the loads and forces given in 5.1 are used in the calculation methods given in 5.2 the necessary design criteria /safety factors of 5.4 are achieved.

5.1 Loads and forces

5.1.1 General

Dock levellers wider than 1,25 m shall, in principle, be designed to withstand the passage of forklift and similar industrial trucks. At widths up to 1,25 m, such dock levellers shall be designed to withstand loads resulting from pedestrian traffic and pedestrian operated industrial trucks only.

5.1.2 Breakdown of loads and forces

5.1.2.1 General

When calculating dock levellers all relevant loads and forces subdivided into primary and supplementary loads and forces shall be taken into account.

Primary loads and forces are as follows:

- dead loads;
- rated load including dynamic effects.

Supplementary loads and forces are considered to be:

- wind forces;
- braking and starting loads;
- inertia forces;
- impact loads.

5.1.2.2 Dead loads

The masses of all permanently fixed and movable components shall be taken as dead loads.

5.1.2.3 Rated load including dynamic effects

In addition to the rated load dynamic effects shall be considered.

The dynamic effects of the rated load due to its movement shall be calculated by multiplying the rated load by the dynamic factor $\varphi = 1,4$. This load shall, depending on the width b of the deck structure, generally be expressed in terms of

- point load for $b \leq 1,25$ m,
- axle load on one axle for $b > 1,25$ m.

Point loads shall be taken as acting over a 150 mm x 150 mm contact area. Axle loads shall be taken as acting over two 150 mm x 150 mm contact areas at 1 m lateral centre distance. These areas shall only apply if the actual conditions do not call for more severe loading.

5.1.2.4 Wind forces

Dock levellers exposed to the environment shall be designed to withstand a wind pressure of $w = 1{,}0\ kN/m^2$ in both directions, directly on to the front of the deck and also directly on to the back of the deck. This is particularly valid for dock levellers which are stowed in the vertical position when not in use.

5.1.2.5 Braking and starting loads

The braking and starting loads for transport equipment operations shall be deemed to be effective at the level of the bridge deck. They shall be calculated at 30 % of the rated load.

5.1.2.6 Inertia forces

The effects of inertia forces shall be taken into account for shock-inducing speed changes of movable components, in particular when a dock leveller travels at operational speed to the final end position in either direction or when the automatic safety device (see 6.3.1) is operated.

5.1.2.7 Impact loads

For vertically stowed and latched dock levellers, a directly applied horizontal load of 1 kN/m of width evenly distributed on a line shall be deemed to be effective at a location 1 m above the top of the hinge line of the dock leveller and at right angles to the bridge deck. For dock levellers shorter than 1 m this horizontal load shall be applied to the top edge of the raised dock leveller. Vehicle impact loads need not be considered.

The defined horizontal load has to be considered, to avoid the impact by a person or manually operated transport equipment from knocking down the vertically stowed dock leveller.

5.2 Calculation of dock levellers

5.2.1 General

Calculations shall be performed in accordance with generally accepted rules.

The source, if generally accessible, shall be stated in case of any unusual formulae or calculation methods. Otherwise formulae shall be developed so that they may easily be verified.

5.2.2 Supporting structures, general

Safety in terms of stress shall be proved by calculation for all supporting structures. The principal dimensions, cross sections, materials and

fastening means shall be stated. The severest individual primary and supplementary load ratings shall be given respectively for each most unfavourable position.

The hinged or telescopic lips, when not supported on the vehicle, need not be calculated with the rated load for the operation of the automatic safety device or the emergency stop device.

All possible load case combinations shall be taken into account.

Adequate strength shall be proved by calculation for components under compression.

5.2.3 Bridge deck

For calculation purposes, only one bearer shall be taken as supporting the point load or the axle load unless lateral distribution of the load is proven.

Where a point load or axle load will be supported by several bearers, such lateral distribution shall be demonstrated for the worst load case (e.g. loading of the end bearer). Non-stiffened sheets shall be calculated according to the plate theory.

Distortion of bridge decks occurring under operating conditions, e.g. by a fork lift truck offset to one side shall be considered.

5.3 Design Test

As an alternative to calculations, components may be subjected to testing to demonstrate their satisfactory design. The rated load multiplied by the dynamic factor (see 5.1.2.3) and the valid safety factor (see 5.4.2) shall be applied to be effective at the most unfavourable position. Components under buckling shall have the safety factor 2,0 applied. There shall be no permanent deformation.

The tests shall be performed and controlled respectively by competent staff or testing organizations.

5.4 Design criteria (factors)

5.4.1 Stability

The stability of dock levellers when subjected to loads specified at 5.1.2 shall be proved by calculation. The effect of the restoring forces shall be 1,5 times the effect of the tipping forces.

5.4.2 Safety in terms of stress

General safety in terms of stress of the steel shall be proved by calculation by applying the uniform safety factor S = 1,5 against the yield stress. Other materials shall be dealt with similarly.

In the load cases where primary and supplementary loads are combined, the safety factor S = 1,33 may be used. This load case is given e.g. for forklift trucks, where the rated load on one axle only occurs when the truck is braked.

5.4.3 Buckling

Proof of adequate resistance to buckling for components subjected to compression shall be obtained in accordance with valid standards and guidelines.

5.4.4 Stresses caused by the operation of the automatic safety device

For this load case the yield stress may be applied (safety factor 1,0) throughout the full cross section as long as there is equilibrium between the internal and external forces.

Proof of resistance to buckling for components subjected to compression may in this case be demonstrated by applying the safety factor S = 1,33.

NOTE: In this load case permanent deformation is acceptable, provided failure of the structure does not occur.

6 Safety requirements and measures

6.1 General safety requirements and measures

6.1.1 Gradients

Dock levellers shall be so designed that the gradient in the operational position does not exceed ± 12,5 % (approximately ± 7°).

6.1.2 Anti slip properties (slip resistance)

The areas of dock levellers which are used by persons or transport equipment shall have anti slip properties. The surface should be so designed that fluids can run away.

NOTE 1: Suitable anti slip properties are surfaces with profiles or anti slip treatment.

NOTE 2: A CEN working group will elaborate an approved test method for slipperiness. The results will be used in a future revision of this standard.

6.1.3 Width

6.1.3.1 Standard width

The dock leveller shall be a minimum of 0,70 m wider than the track width of the transport equipment as specified in the manufacturer's instruction, but at least 1,25 m wide.

NOTE 1: This is to make available a minimum safety width on each side of transport equipment.

NOTE 2: The width of a dock leveller should correspond to the width of the loading surface of the loading vehicle.

6.1.3.2 Reduced width

As an exception the minimum width of 1,25 m may be reduced if:

- facilities at the operator's premises make this necessary and the operator takes measures to prevent persons or transport equipment falling,
- transport equipment is guided onto the dock leveller and the operating personnel can use an area in the middle of the leveller,
- dock levellers are used for access to doors in vehicles or buildings which are less than 1,25 m in width and precautions against falling have been taken (e.g. the gap between the dock and vehicle is less than 0,20 m).

6.1.4 Support in the resting position

Dock levellers which are attached to ramp edges and folded upwards in the unused position shall be equipped with automatically operating and positively engaged mechanical devices to prevent them from falling down, e.g. hooks, bolts or bars.

These devices shall withstand the load resulting of the dead load of the dock leveller in the stowed position, of the wind force as given in 5.1.2.4 and of the impact load given in 5.1.2.7. The device shall be so designed that it can not be disengaged unintentionally.

Dock levellers which are integrated into traffic areas shall be so designed that they form a level surface in the resting position with their neighbouring surfaces and shall be automatically supported to accept the rated load.

6.1.5 *Tripping hazards*

Dock levellers shall be designed so that they do not cause a tripping hazard in the traffic area for loading.

Dock levellers, supported on the loading surface of a vehicle, shall automatically adjust to height changes (free floating condition) during the loading operation.

Tripping hazards between dock levellers and vehicle loading surfaces which occur if the vehicle loading surface is tilted along its longitudinal axis shall be prevented, for example by means of

- sufficient torsional flexibility of the dock leveller

or

- hinged lips which automatically adapt themselves to the lateral angle of the vehicle loading surface.

6.1.6 Prevention of an unsupported operational position

Manual operated dock levellers which are installed in traffic areas shall be designed so that, when unloaded, they cannot remain in the near horizontal position unless they are supported. A difference in height of max 100 mm to the supported position, measured at the front edge, is permitted.

A near horizontal position is a position with a maximum vertical distance of 50 mm between dock plate and the level of the traffic area, measured at the front edge of the dock leveller.

The supporting device, e.g. blocks, bars or bolts shall be able to withstand the rated load without permanent deformation.

6.1.7 Prevention of unintentional movement

Dock levellers shall be designed to prevent unintentional movement during the loading operation.

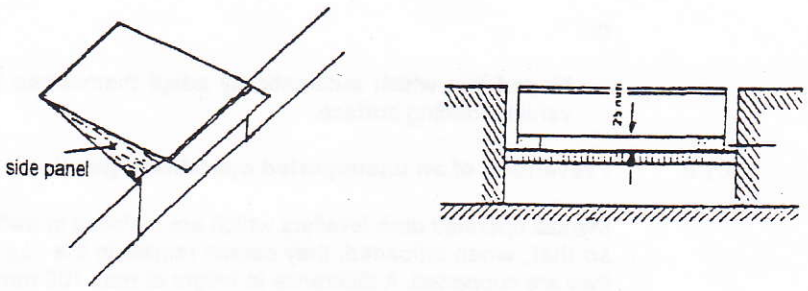
If power driven dock levellers are equipped with the function "automatic return to resting position", the movement to the resting position shall only be started when the vehicle has left the loading position.

Manually operated dock levellers shall be so designed that they cannot leave their supported position on the vehicle during the loading process.

6.1.8 Crushing and shearing hazards

Crushing and shearing hazards at dock levellers, between dock levellers and neighbouring components and between dock levellers and the vehicle loading surface shall be avoided or, if unavoidable, shall be safeguarded.

Crushing and shearing hazards between the edges of installed dock levellers and the neighbouring traffic areas can be avoided by using rigid side panels (see figure 3 a).



a) Side panels to avoid crushing of feet

b) Safety distance between lip edge and frame

Figure 3: Safeguarding of crushing and shearing hazards

Crushing and shearing hazards between the folded lip and the dock leveller frame are to be considered as being safeguarded if a vertical or horizontal safety distance of at least 25 mm between the lip edge and the frame components is maintained when supported (safety distance for fingers according to EN 349:1993). Support blocks of the required static width require no safety distance (see figure 3 b).

Crushing and shearing hazards between dock levellers and vehicle loading surfaces are to be considered as safeguarded

- for manually operated dock levellers by the operating person,
- for power operated dock levellers by correspondence with the requirements of 6.3.2.2.

6.1.9 Supporting device for maintenance

Installed dock levellers shall be fitted with a mechanical prop or device which, when operated, will support the unloaded leveller at its maintenance position and allow maintenance work to be carried out in safety. This prop or device shall form part of

the dock leveller and it shall not be possible for it to be unintentionally disengaged.

6.1.10 Visibility of lifted or lowered dock levellers

The visible side components of installed dock levellers, when raised, and the visible side areas when at lowered position as well as the side components of dock levellers resting in a vertical position shall be clearly marked in yellow/black to indicate danger. The lips are excluded from this requirement.

6.2 Additional requirements for manually operated dock levellers

Manually operated dock levellers which are moved by directly applied human forces shall fulfill the requirements of 6.2 in addition to 6.1.

6.2.1 Protection against dangerous movements in use

Dock boards shall be designed so that they can be secured and supported so that they will not move in an unsafe manner when used by pedestrians or vehicles.

This may be achieved by bolts, bars or teeth which engage at the dock edge or the frame of the vehicle.

6.2.2 Manual operation

Dock boards to be carried by one man shall not exceed 300 N in weight. If two men are to be used then the weight shall not exceed 600 N.

All dock boards of over 600 N in weight shall be equipped with devices that allow the handling of the dock boards by using forces lower than 300 N, or shall be fitted with means to enable safe movement by mechanical handling equipment.

Devices, e.g. castors, to lighten the movement and operation of dock boards are recommended.

Lifting handles shall be fitted to all dock boards to be manually moved.

Dock boards which are hinged at dock edges shall not require a manual force of more than 300 N to pivot them.

Mechanically operated dock levellers installed in docks or in the floor shall not require a manual force of more than 300 N to operate them.

6.2.3 Adjustments of dock levellers with mechanical drives

All counterbalanced, spring operated and other stored energy operated dock levellers shall be provided with means of adjustment to compensate for wear and frictional changes which could affect the balance and the bias of the platform.
The adjustment device shall ensure that the manual operating force remains below 300 N.

6.3 Additional requirements for power-operated dock levellers

6.3.1 Automatic safety device

Unsupported power operated dock levellers shall be secured automatically against uncontrolled and dangerous lowering when carrying more than 10 % of the rated load, e.g. by a hydraulic valve mounted directly on the lifting cylinder.

In the cases of a dock leveller with a small closed height due to the lack of space on the site, if the lifting cylinder has to be inclined at a steep angle, the 10 % value may be increased up to 25 %. This value shall be specified by the manufacturer.

If such a lowering starts, e.g. by the loading vehicle driving away before the dock leveller has returned to its resting position, the dock leveller shall either

- stop within a distance of 0,20 m, measured at the front edge of the bridge deck

or

- reduce the lowering speed to not more than 0,05 m/s.

An automatic safety device is not necessary if the vehicle is interlocked with the dock leveller during the loading process, e.g. by a hook.

6.3.2 Requirements for control

6.3.2.1 Control system

The control system shall be designed according to ~~prEN 954-1:1996~~, category 2.

6.3.2.2 Location of controls

The control panel of power-operated dock levellers shall be located and designed so that operating personnel can easily observe all movements and the operator is clear of all risks from the leveller and the vehicle.

6.3.2.3 Controls

Controls shall be designed so that if they are released

- the dock leveller will become stationary,

or

- the dock leveller will lower itself under its own weight at a speed not exceeding 0,15 m/s, measured at the front edge of the dock plate.

6.3.2.4 Control of automatic return

In exception to 6.3.2.3 the return movement of the dock leveller to its resting position may be automatic, started automatically or by the operator.

Automatic return is only allowed when the dock leveller is unloaded and when the vehicle has left its loading position.

Unintentional movement shall be avoided according to 6.1.7.

6.3.2.5 Design of controls

The controls of power-operated dock levellers shall be arranged and marked so that their functions are easily understandable. They shall also be designed, arranged and safeguarded against unintentional operation, e.g. by flush installation on the control panel.

NOTE: The use of symbols or pictograms is recommended.

6.3.2.6 Emergency stop device

Power-driven dock levellers shall be equipped with a clearly marked and easily accessible emergency stop device according to EN 418:1992 by which all movements of the dock leveller can be stopped in the event of danger. The emergency stop device shall be located on the control panel or close by.

The function of the emergency device and the main switch may be combined in one switching unit which fulfills the following requirements for both:

- power-switching performance (AC 1)
- one ON and OFF position only
- lockable in the OFF position

and

- easily visible red control component in front of contrasting yellow background.

6.3.2.7 Main switch

All power-operated dock levellers shall be equipped with a main switch to disconnect all electrical live conductors under full electrical load conditions. The main switch shall be lockable at the Off position to prevent unintentional and unauthorized operation.

6.3.2.8 Stopping of dock levellers

In the event of activation of the emergency switch, main switch, or of a power loss, all movements of power-driven dock levellers shall stop.

6.3.2.9 Prevention of automatic start-up

Power shall be restored after a power loss or the activation of the main switch or emergency stop control - see 6.3.2.8 - by a manually operated control. The control shall ensure that there is no movement uncontrolled by the user of the dock leveller unless this is to enable the dock leveller to return to the resting or operating position at speeds not exceeding 0,15 m/s.

6.3.3 ~~Requirements for electrical equipment~~

6.3.3.1 General

The electrical installation shall be designed and the electrical equipment shall be selected such that avoidance of hazards due to the electrical energy is ensured.

Electrical installations shall be in compliance with, and designed according to, the relevant standards, e.g. HD 384.4.41S1 and HD 384.4.47S1.

The electrical installations shall be capable of withstanding the mechanical, thermic and climatic stresses expected to occur during normal service duties of the dock leveller.

The potential hazards of the electromagnetic characteristics of the installation shall be considered.

The nominal data specified for the electrical equipment by the manufacturer of the dock leveller shall not be exceeded.

6.3.3.2 Protection mode

The minimum mode for the protection of electrical housings from the ingress of water and foreign matter shall be in accordance with IP 54 of EN 60529:1991.

6.3.3.3 Contactors

Contactors shall have a mechanical life compatible with a periodic rating class 0.3 and meet the minimum design requirements to service category AC 3 according to EN 60947-4-1:1991; HD 419.2.

6.3.3.4 Solenoid valves

Coils of solenoid valves shall be designed for continuous operation.

6.3.4 Requirements for hydraulic and pneumatic equipment

6.3.4.1 General

Hydraulic and pneumatic equipment shall be designed according to EN 982:1996 and EN 983:1996.

Hydraulic and pneumatic cylinders, pipes and their connections which may be subjected to the pressure permitted by the pressure relief valve shall be designed to withstand at least

- in hydraulic systems two times this pressure
- in pneumatic systems three times this pressure

without permanent deformation.

Hoses shall be dimensioned in order to withstand a bursting pressure equal to at least three times the pressure permitted by the pressure relief valve.

Parts which are subject to overload under automatic safety operation shall be

designed to withstand this overload pressure.

6.3.4.2 Pressure relief valve

A pressure relief valve shall be fitted in all systems and shall be positioned and set so that it is not possible to operate at a pressure of more than 115 % of the normal working pressure.

This valve shall be secured against unauthorized adjustment.

6.3.4.3 Cylinders

It shall not be possible for the positioning mechanism to lower away from the platform if the platform travel is unintentionally blocked.
It shall not be possible for the supporting medium to drain out of the cylinder if the platform descent is blocked.

Cylinders shall be mounted so that they are subjected only to axial loads.

6.3.4.4 Venting

When a hydraulic system is fitted, unless it is designed to be self venting, means shall be provided for venting trapped air which could interfere with the safe operation of the dock leveller.

6.3.4.5 Pipes and hoses

Hydraulic and pneumatic pipes and hoses shall be designed, mounted and (where necessary) protected to preclude damage from rubbing, chafing, bending etc. during normal operation.

6.3.4.6 Air supply

When a pneumatic system is fitted, the air supplied to the dock leveller shall be clean and free from liquids to the manufacturer's specification.

dia-hinges - 50mm.
dia-lifting cylinder - 50mm
Uprotation cylinder - dia - 63

Hinges - 50mm
lifting cylinder - 50mm.
lip cylinder - 63mm

5003 blue

7 Verification

Table 2 indicates the method(s) by which the safety requirements and measures described in clause 6 shall be verified, together with a reference to the corresponding subclauses in this standard.

Table 2: Means of verification of the safety requirements and measures

Subclause	Safety requirements and measures	Inspection	Function test	Measurement
6.1.1	Gradients			x
6.1.2	Anti slip properties	x		
6.1.3	Width			x
6.1.4	Support in the resting position	x	x	
6.1.5	Tripping hazards	x	x	
6.1.6	Prevention of unsupported positions		x	
6.1.7	Prevention of unintentional movement		x	
6.1.8	Crushing and shearing hazards	x		x
6.1.9	Supporting device for maintenance	x	x	
6.1.10	Visibility of lifted or lowered dock levellers	x		
6.2.1	Protection against dangerous movements in use		x	
6.2.2	Manual operation	x		x
6.2.3	Adjustments of dock levellers with mechanical drive	x		
6.3.1	Automatic safety device		x	
6.3.2.1	Control system		x*)	
6.3.2.2	Location of controls	x		
6.3.2.3	Controls		x	
6.3.2.4	Control of automatic return		x	
6.3.2.5	Design of control instruments	x		
6.3.2.6	Emergency stop device		x	
6.3.2.7	Main switch		x	
6.3.2.8	Stopping of dock levellers		x	
6.3.2.9	Prevention of automatical start-up		x	
6.3.3.1	Requirements for electrical equipment; General	x		
6.3.3.2	Protection mode		x*)	
6.3.3.3	Contactors	x		
6.3.3.4	Solenoid valves	x		
6.3.4.1	Requirements for hydraulic and pneumatic equipment; General		x	
6.3.4.2	Pressure relief valve		x	
6.3.4.3	Cylinders		x	
6.3.4.4	Venting	x		
6.3.4.5	Pipes and hoses	x		
6.3.4.6	Air supply	x		

*) or declaration of the manufacturer

8 Information for use

Manufacturers, importers or suppliers shall supply dock levellers with the essential marking and with written instructions for use including installation, operation, maintenance and inspection in the official language(s) of the country of installation.

8.1 Marking

The following data shall be easily visible and permanently fixed to each dock leveller:

- manufacturer or supplier,
- country of the manufacturer or supplier,
- type,
- manufacturing number,
- year of manufacture,
- rated load

and additionally for dock levellers with external power supplies:

- details of supply required (for electrical equipment: voltage, frequency, power).

8.2 Instructions for use

8.2.1 Installation instructions

All details to ensure an easy and safe installation shall be given by manufacturers, importers or suppliers, e.g. power connection, dimensions for the pit, fixing details.

8.2.2 Operating instructions

The instructions shall contain necessary details about safety in use, regarding

- purpose of the equipment,
- initial start-up,
- use and function during operation and faults in operation,
- essential instructions for safe operation.

The essential instructions for safe operation shall particularly contain, but not be restricted to, the following:

- Dock levellers shall not be operated above their rated load.
- Dock levellers shall not be operated above their permitted gradients.

- The vehicle shall be positioned not more than 0,20 m from the loading dock if dock levellers with a reduced width as given in 6.1.3.2 are used.
- Transport equipment with a track width of more than the permitted value shall not be used.
- When not in use dock levellers shall be immediately returned to their resting position.
- Unless an automatic return system is fitted, care shall be taken to ensure that the vehicle does not depart before the dock leveller is returned to its resting position.

- Dock boards shall be fixed during the operation so that they are secured against unintentional movement.
- Dock boards shall be safely stored after use.

8.2.3 Brief operating instructions

Dock levellers shall be equipped with brief operating instructions in the same language(s) as described in clause 8 or with illustrations which can be permanently affixed and easily visible near the operators location. These instructions shall include the rated load.

8.3 Instructions for maintenance and inspection

In order to ensure proper and safe maintenance and inspection of the dock leveller and if necessary special devices, the user shall comply with the instructions on maintenance and inspection provided by the manufacturer. Where applicable, this shall include, but not be limited to, the following.

8.3.1 Maintenance

Maintenance work shall be carried out by a competent person at regular intervals in accordance with the manufacturer's documentation. Existing regulations and safety requirements shall be observed.

The drive units of power-operated dock levellers shall be switched off and safeguarded against unintentional and unauthorized operation prior to carrying out service or maintenance. An exception hereto is the function test, where the maintenance person observes the dock leveller carrying out certain functions in order to detect failures.

If service or maintenance work is carried out underneath raised dock levellers they are to be secured against descending by use of the maintenance prop or device (see 6.1.9).

8.3.2 Inspection

Dock levellers which are permanently affixed to buildings shall be inspected for safe function by a competent person prior to the initial start-up. Intervals for inspections shall be given by the manufacturer.

Other types of dock levellers shall be inspected regularly as necessary for safety according to manufacturers instruction.

The inspection includes in particular:

- visual inspection regarding outwardly recognizable wear and damage,
- functional checks,
- completeness and effectiveness of safety installations.

Dock levellers shall also be inspected after major repairs, e.g. repairs on structural components by welding. The extent of inspection required is determined by the extent of the repair work needed.

Written records of the results of all inspections with the date, name, address and signature of the competent person shall be kept by the user.

Annex A (informative)

Examples of suitable tests

A.1 General

If a sample of a dock leveller is subject to a type examination, the following points are examined in particular - if relevant:

- completeness and correctness of documentation,
- compliance of the test sample with documentation,
- width and gradient,
- anti slip characteristics of the traffic areas,
- effectiveness of the safety devices to prevent unintentional movement,
- support in resting position,
- absence of tripping hazards,
- effectiveness of safety devices to prevent unintentional lowering,
- effectiveness of safety devices to prevent crushing and shearing hazards,
- effectiveness of the supporting device,
- labels and markings,
- manual operating forces,
- condition and quality of the hydraulic or pneumatic equipment,
- compliance of the electrical equipment with the requirements,
- stability of the dock leveller in accordance with A.2,
- satisfactory function of the automatic safety device in accordance with A.3.

The results of the tests as well as the name and address of the testing persons or institute are recorded in a signed report.

A.2 Stability test

The test load is the rated load multiplied by the safety factor (see 5.4.2) and the dynamic factor (see 5.1.2.3). The dock leveller is supported in a horizontal position on the completely extended lip. The test load is positioned in the center and also offset on the dock leveller in accordance with figure A.1 and figure A.2.

After the test, the dock leveller shall not show a visible permanent deformation

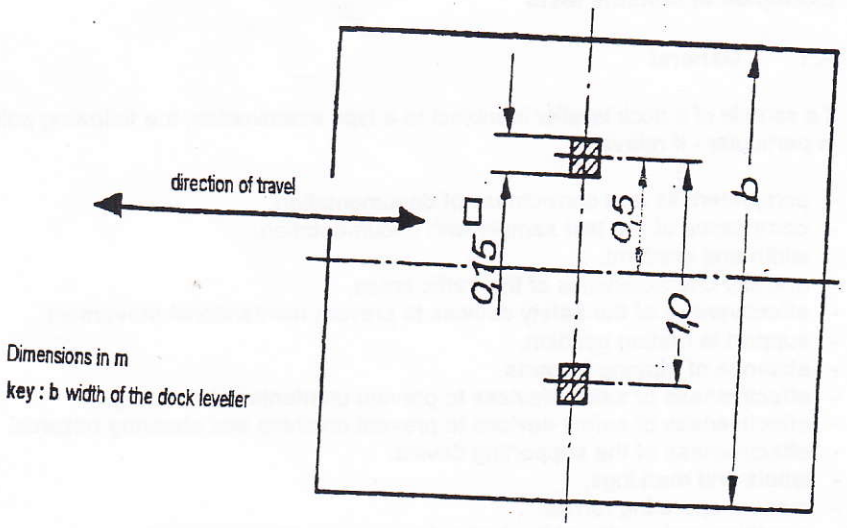


Figure A.1: Stability test for load case 1

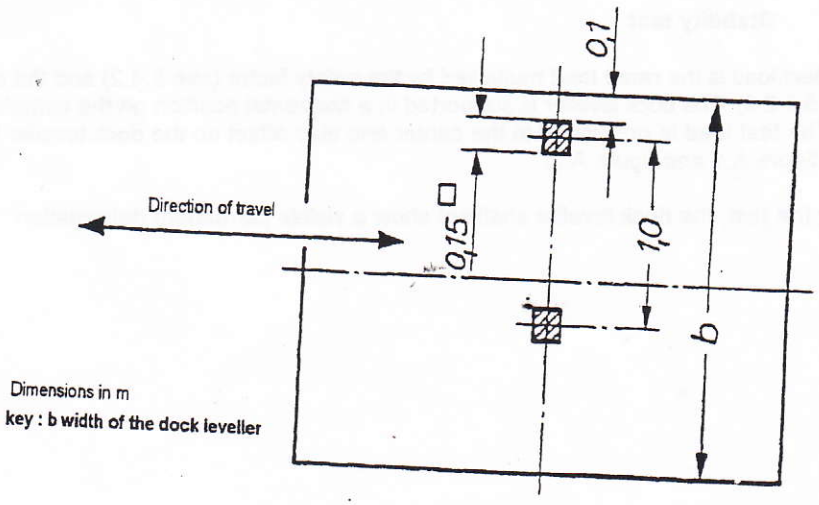


Figure A.2: Stability test for load case 2

A.3 Function test of the automatic safety device

The test load of 10 % (respectively 25 % see 6.3.1) of the rated load is positioned on the dock leveller in the middle of the deck plate. The dock leveller is supported horizontally on the lip which is in the operational position. The temperature of the environment is 20 °C ± 10 °C.

After removing the supporting device the dock leveller shall be stopped or have it's speed reduced by the automatic safety device. The falling distance shall not be greater than 0,20 m, measured at the front edge. The reduced speed shall be not more than 0,05 m/s.

Annex ZA (informative)

Relationship of this European Standard with EU Directives.

This European standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of following EU Directives:

Machinery Directive 89/392/EEC,
its amendments 91/368/EEC and 93/44/EEC,
Low Voltage Directive 73/23/EEC

except 6.2 of this standard.

Compliance with this standard provides one means of conforming with the specific essential requirements of the Directives concerned.

WARNING: other requirements and other EU Directives may be applicable to the products falling within the scope of this standard.